

50.007 Machine Learning

Dimensionality Reduction with Autoencoders

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Learning Objectives ~~*~~ Not Tested in Exam

You should be able to know:

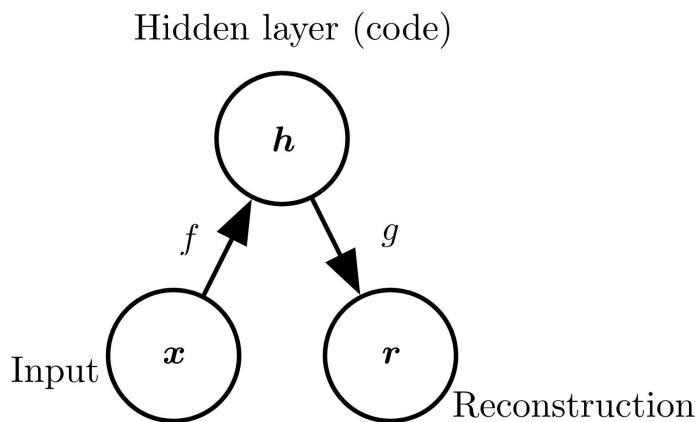
- Autoencoder architecture
- How Autoencoder reduce dimension of data

This is a filler episode - not tested!



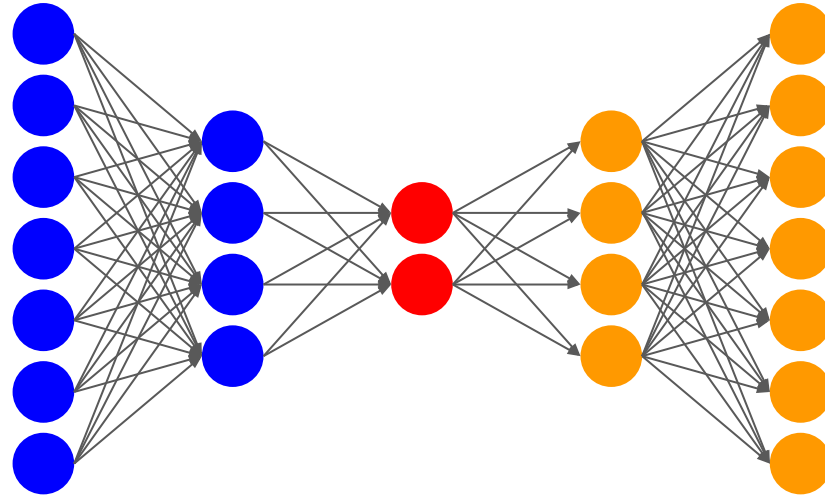
What is Autoencoder?

- An unsupervised learning approach that adopts a deep learning neural network architecture
- Perform **representation learning**



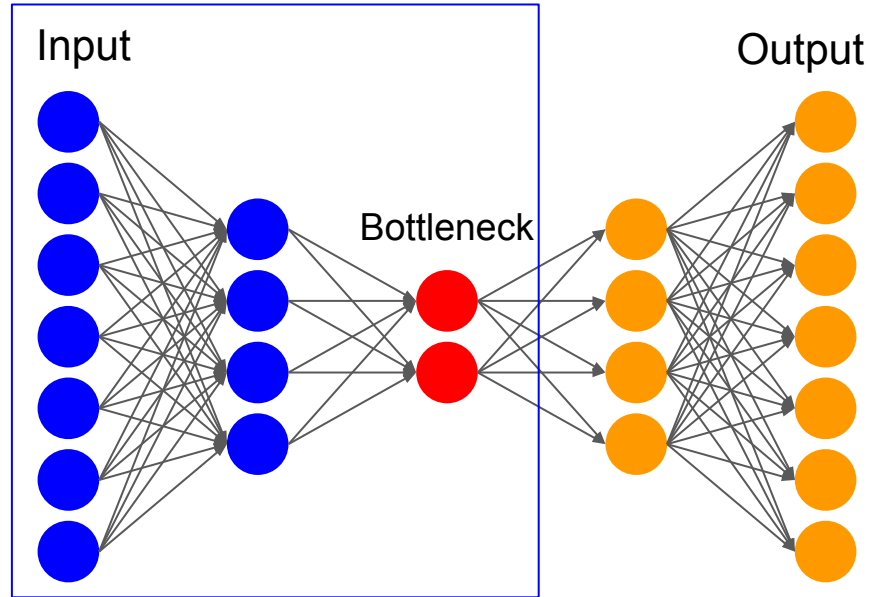
Autoencoder Architecture

- Key Idea: Create an architecture with a **bottleneck** which captures the **low-dimensional representation** of the original input data



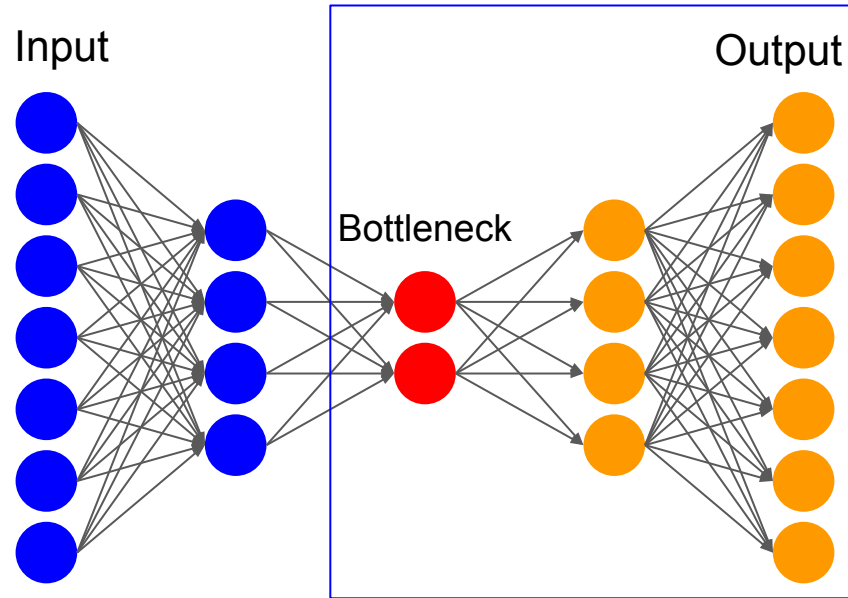
Autoencoder Architecture

- **Encoder:** Compress data into lower dimensional representation



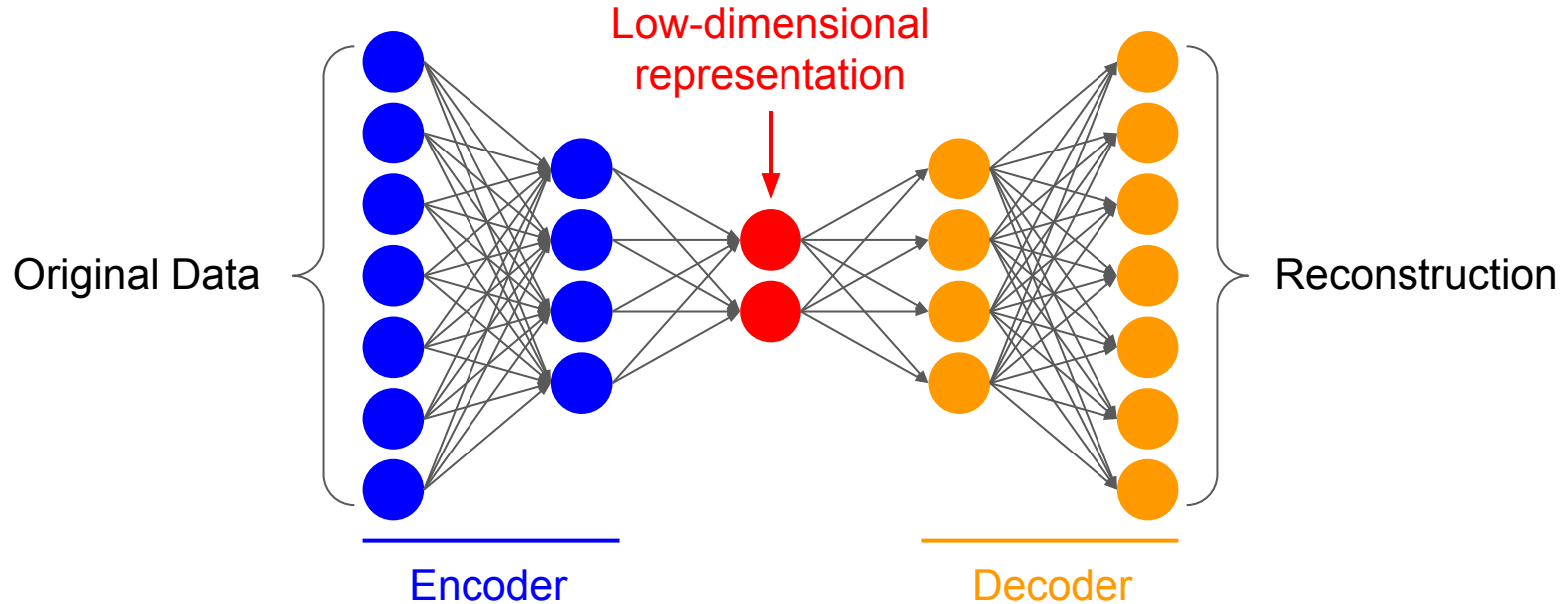
Autoencoder Architecture

- **Decoder:** Decompress representation back original data (reconstruction)



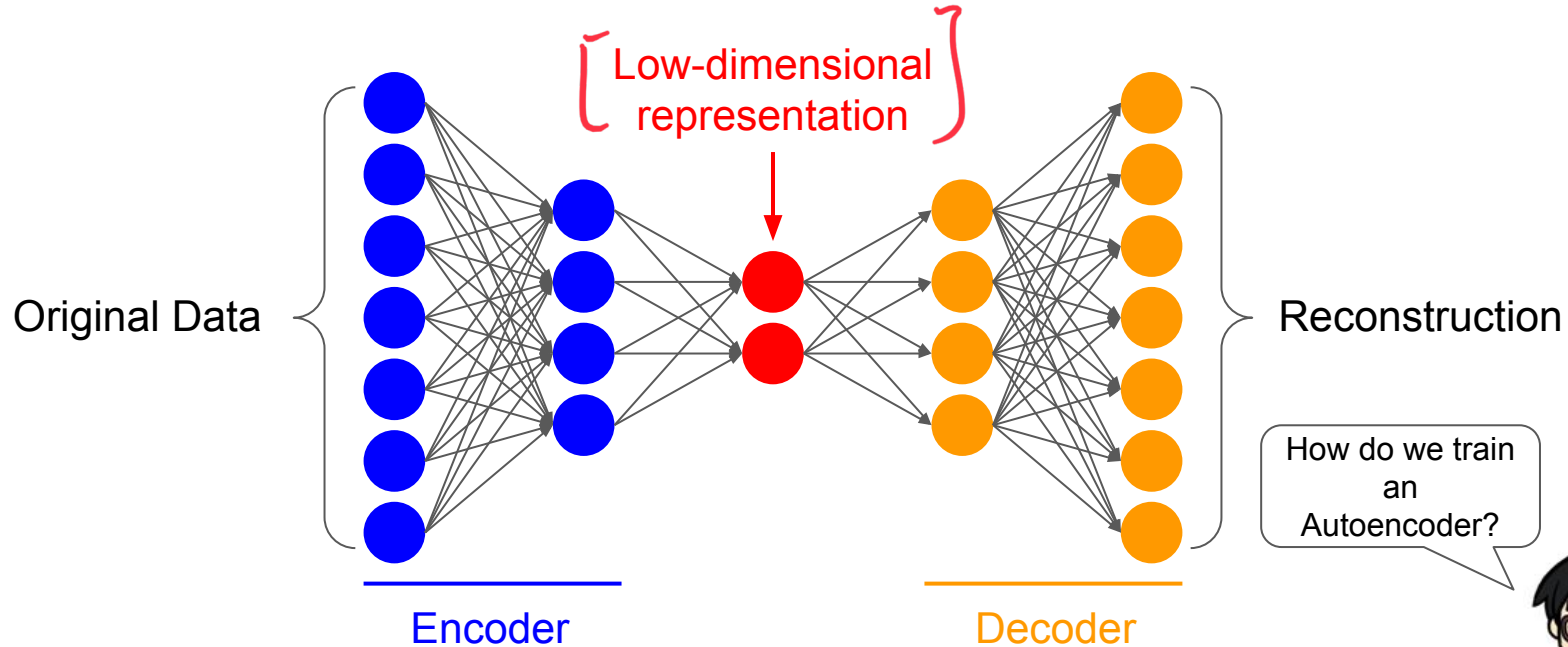
Autoencoder Architecture

- Autoencoder = Encoder + Decoder



Autoencoder Architecture

- Autoencoder = Encoder + Decoder



Training Autoencoder

- Backpropagation
- Minimize Reconstruction Error
 - L is a loss function penalizing $g(f(x))$ for being dissimilar from x , such as the mean squared error

Loss Function $\longrightarrow L(x, \underbrace{g}_{\text{Decoder}}(\underbrace{f}_{\text{Encoder}}(x)))$

Optimal Autoencoder

- Sensitive enough to input data to reconstruct it

$$L(x, g(f(x)))$$

Optimal Autoencoder

- Sensitive enough to input data to reconstruct it
- Insensitive enough to input data not to overfit it (i.e., just copy the input)

$$L(x, g(f(x)))$$

Optimal Autoencoder

- Sensitive enough to input data to reconstruct it
- Insensitive enough to input data not to overfit it (i.e., just copy the input)

makes x very similar to the input
but NOT the same!

$$L(x, g(f(x))) + \text{regularization}$$



variational autoencoder (VAE)

Autoencoder Applications

- Dimension Reduction *
- Generation
- Denoising
- Many more!

To know more, take the
deep learning class!

PCA vs Autoencoder

- Both perform dimensionality reduction
- PCA learns linear relationship → no matter how many lines it draws.
- Autoencoder learns non-linear relationship*
- Autoencoder = PCA, if it uses linear activation functions

*Use non-linear
activation
function!



What you should know?

- Autoencoder architecture
- How Autoencoder reduce dimension of data

Suggestion: If you're not very clear or want to learn more, please do read: "Goodfellow, Ian, Yoshua Bengio, and Aaron Courville. [Deep learning](#). MIT press, 2016." It will help you to understand the concept.

Acknowledgement

- *The following material have been referenced or partially used:*
 - *Goodfellow, Ian, Yoshua Bengio, and Aaron Courville. Deep learning. MIT press, 2016.*